



Name _____ HR _____

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7th Grade Science Unit 6: Thermal Energy STAR Cards!

QUESTION	ANSWER
6a)  Everything in the world is made of <u>atoms</u> that vibrate. 1. What would happen to the vibration of <u>atoms</u> of a door if it was heated up? 2. What would happen to the vibration of the <u>atoms</u> of a spoon if it was cooled down?	6a) 1) The door's atoms would vibrate faster 2) The spoon's atoms would vibrate slower
6c)  1. Define the following ● Kinetic Energy ● Thermal Energy ● Temperature 2. Describe how they are related to heat energy	6c) 1. ● <u>Kinetic Energy</u> is the amount of movement in an object's particles ● <u>Thermal Energy</u> is the total amount of kinetic energy in an object ● <u>Temperature</u> is the average kinetic energy in an object. 2. As heat flows through an object, the kinetic energy, temperature and thermal energy increase

QUESTION	ANSWER
6d) ☆☆☆ 1. If you put a cold spoon in a cup of hot coffee, explain what would happen to the temperature of both the coffee and the spoon.  2. What is this process called in science?	6d) 1. The heat would continue to flow from the hot coffee to the cold spoon until they both reach equal temperatures. 2. This is called reaching thermal equilibrium.
6e) ☆☆☆ Define the 3 types of heat transfer: ● Conduction ● Convection ● Radiation	6e) ● <u>Conduction</u>: Heat is transferred by direct contact ● <u>Convection</u>: Heat is transferred through the movement of liquids or gasses ● <u>Radiation</u>: Heat is transferred by waves through empty space
6f) ☆☆☆ What is thermal conductivity?	6f) Thermal Conductivity is an object's ability to transfer heat (For example, a metal spoon has high thermal conductivity, but a wooden spoon has low thermal conductivity.)